Revision History

Model	Revision	Date	Description
SSDP3A0	A	2025-07	Concept design using the LM70880 buck converter and LM5069-1 hotswap controller. Abandoned due to supply chain issues and cost constraints.
SSDP3A1	A	2025-08	Concept design using the LM5146 buck controller and TPS2490 hotswap controller. Abandoned due to space and cost constraints.
SSDP3A2/SSDV3	A	2025-09	Concept design using the LM5146 buck controller in a custom 1/8th brick form factor. Accepted design for revision.
SSDV3	B	2025-10	Concept design using the cheapest parts available on LCSC. Limited part availability and NRND parts made future production of this revision difficult.
SSDV3	C	2025-10	Prototype design featuring reverse polarity protection in exchange for a smaller TVS diode on the input. Smaller, active-production 3x3mm PQFN-8 FETs used in place of SO-8 FETs. Used for preliminary integration testing and validation.
SSDV3	D	2025-11	Production revision from Rev. C., addressing issues found in testing: - Fixed electrolytic input capacitor unprotected from reverse polarity. - Optimized FETs for Qoss + Qrr losses. - Replaced RPP ground diodes with 100V rated diodes. - Revised capcaitor selection for higher temperature rating: X5R -> X7R. - Stricter adherence to voltage clearance requirements. - Refactored schematic for A4 paper. - Removed 1210 capacitors for 1206 to reduce vibration sensitivity.

rpp

File: rpp.kicad_sch

buck_converter

File: buck_converter.kicad_sch

Heatsink

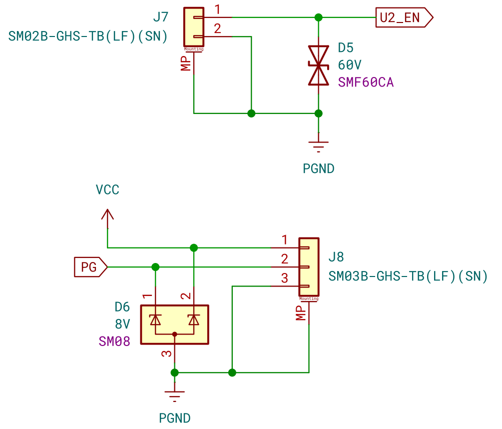
HS1
241802B92200G

JLCPCB Eco PCBA Tooling Holes

H1
MountingHole
H2
MountingHole
H3
MountingHole

CAUTION
HOT SURFACE

Mechanical



Auxiliary I/O

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A 12S battery eliminator circuit (BEC) for UAS applications.

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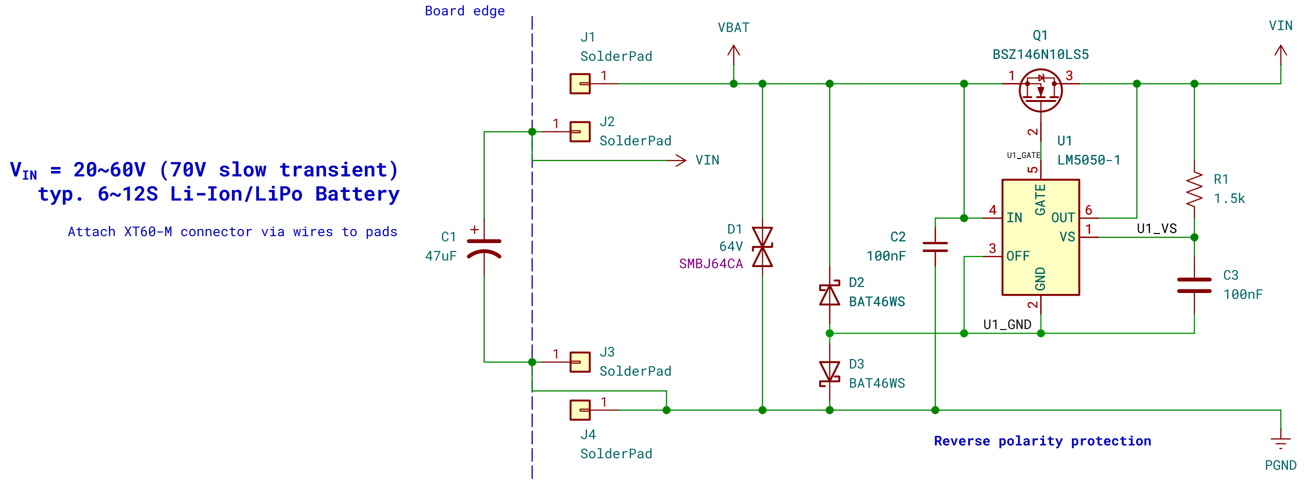

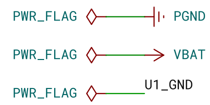

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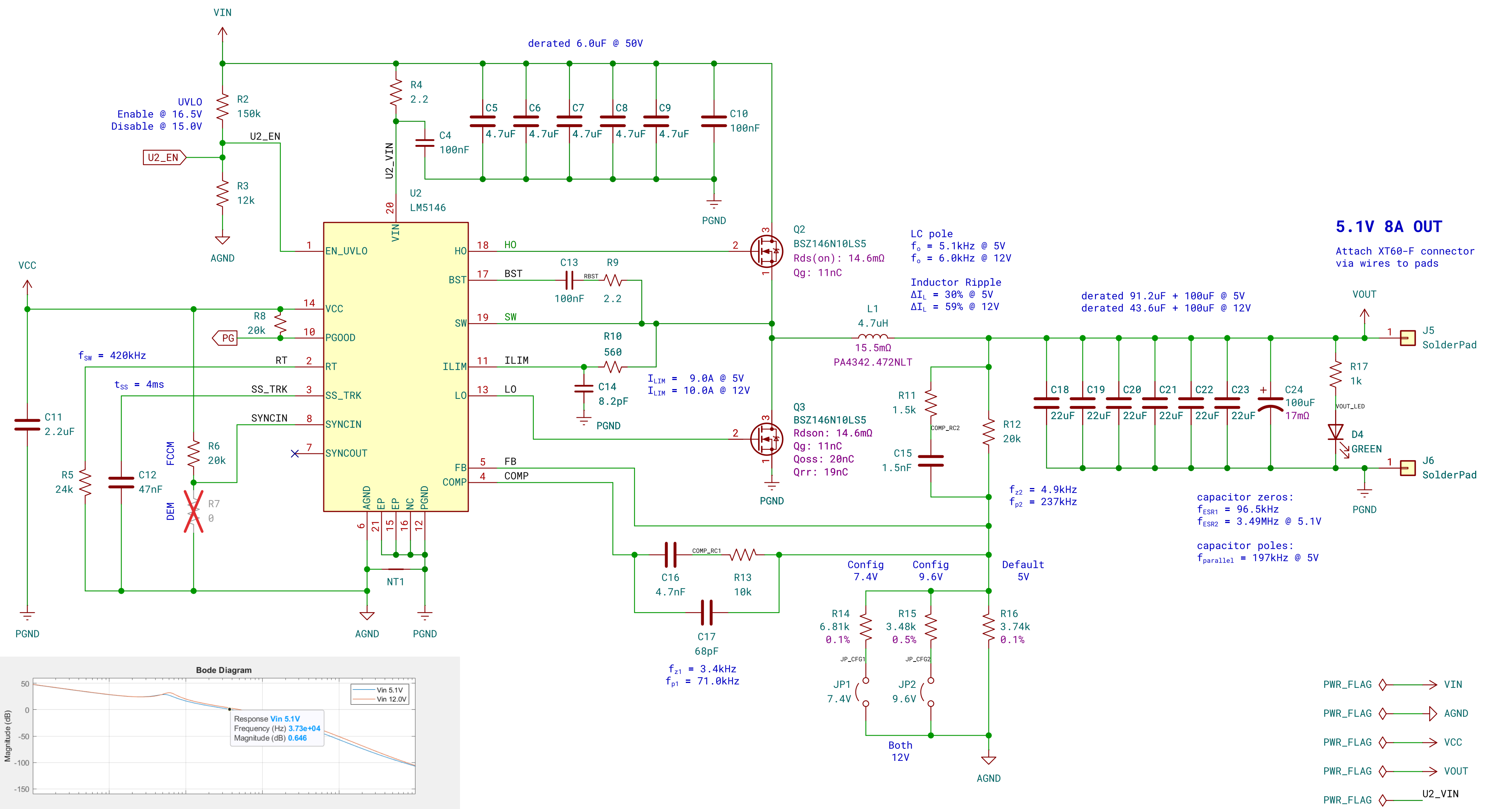

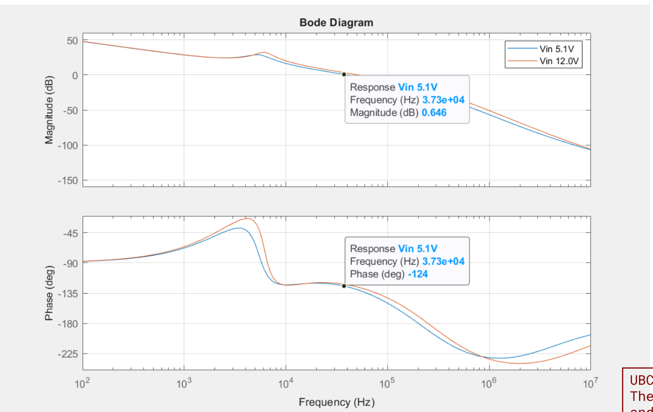

Open Loop Characteristics

A 12S battery eliminator circuit (BEC) for UAS applications.

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Sheet: /buck_converter/
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